

### Demonstration of Proposed Features

|                      |                                                                                        |
|----------------------|----------------------------------------------------------------------------------------|
| <b>Date</b>          | 2 July, 2026                                                                           |
| <b>Team ID</b>       |                                                                                        |
| <b>Project Name</b>  | Hydra Shield: An Intelligent Flood Prediction and Early Warning System (Rising Waters) |
| <b>Maximum Marks</b> | 1 Mark                                                                                 |

### Demonstration of Proposed Features:

This document captures all the features proposed during the project planning phase and verifies whether they were successfully implemented and demonstrated in the HydraShield – Intelligent Flood Prediction and Early Warning System.

| S.No | Feature Name                | Description                                                                         | Status<br>(Implemented /<br>Partial /<br>Pending) | Demonstrated<br>(Yes / No) | Remarks                                      |
|------|-----------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------|----------------------------------------------|
| 1    | Environmental Data Input    | Accepts weather and environmental parameters through a user-friendly web interface. | Implemented                                       | Yes                        | Successfully accepts user inputs.            |
| 2    | Data Preprocessing          | Cleans and preprocesses the input data before prediction.                           | Implemented                                       | Yes                        | Improves model performance.                  |
| 3    | Machine Learning Prediction | Predicts flood risk using the trained machine learning model.                       | Implemented                                       | Yes                        | Random Forest model integrated successfully. |
| 4    | Prediction Result Display   | Displays the predicted flood risk and safety                                        | Implemented                                       | Yes                        | Provides real-time prediction results.       |

|   |                                  |                                                                                           |             |     |                                                |
|---|----------------------------------|-------------------------------------------------------------------------------------------|-------------|-----|------------------------------------------------|
|   |                                  | recommendations instantly.                                                                |             |     |                                                |
| 5 | Model Performance Evaluation     | Evaluates the trained model using Accuracy, Precision, Recall, and F1-Score.              | Implemented | Yes | Performance metrics evaluated successfully.    |
| 6 | Flask Web Application Deployment | Integrates the trained model into a Flask-based web application for real-time prediction. | Implemented | Yes | Successfully demonstrated end-to-end workflow. |

### Feature Implementation Summary

| Feature                         | Value       |
|---------------------------------|-------------|
| Total Features Proposed         | 6           |
| Total Features Implemented      | 6           |
| Total Features Demonstrated     | 6           |
| Overall Implementation Rate (%) | <b>100%</b> |